Ahmedabad University

CSE623 Machine Learning

Weekly Report 3

Section 1

Submitted to Faculty: Prof. Mehul Raval

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Student Details

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1. Objective Task for This Week

The objective task for this week was to continue working on the previously received dataset that the team attempted to preprocess accurately. The team wanted to move forward with the accurate preprocessing and focus on trajectory and motion analysis and visualization. Since the vehicle annotation using Oriented Bounding Box the team did before was incorrect, the team initially had to correct that before we could extract useful information from trajectory data.

Our goals for this week were:

- Fixing the incorrect Oriented Bounding Box annotations
- Visualization of position of vehicles in frame 1 using the extracted information from OBB annotations.
- Visualization of vehicle velocities to understand motion patterns along with comparison with smoothed and original velocity to determine which is better for identifying the stopping behaviour of vehicles.

2. Completed Tasks

This week, the work was split into 2 main tasks- OBB Generation and Velocity Visualization.

In OBB generation, the initial step consisted of extracting the first row in each of the CSV files of the vehicles. The first row had the earliest known position of the vehicle hence, the team used that to acquire the initial information on the bounding box. The extracted rows were bundled together and used to create the Oriented Bounding Boxes of the first frame, and this is used to visualize the position and the orientation of vehicles that are at the start of the sequence.

Simultaneously, velocity visualization was also being worked on by the team. Each vehicle is measured twice in terms of velocity- original velocity and smoothed velocity. The data set includes two velocity values; original velocity and smoothed velocity. Original velocity is the raw calculated speed of the vehicle on a frame by frame basis and smoothed velocity is a filtered version which removes noise, sudden changes in the data. Both of them were used for analysis to decide which one of them gives a better picture of vehicle movement and can be more effectively used in recognizing the patterns of stops such as hard stops or momentary stops.

Their behavior was analyzed by plotting graphs with help of two velocity data. The objective of developing such plots is to compare the trends of the two types of velocities and get a better idea on which one offers more accurate information. By evaluating the change in velocity over time in these graphs, the team can get a decide which is more suitable for detecting hard and momentary stops in vehicle trajectories.

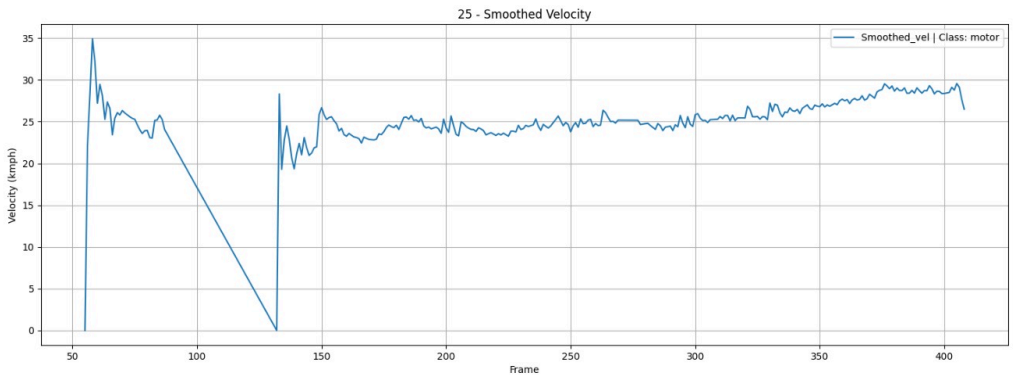

Fig. 1: Example of Monetary Stop

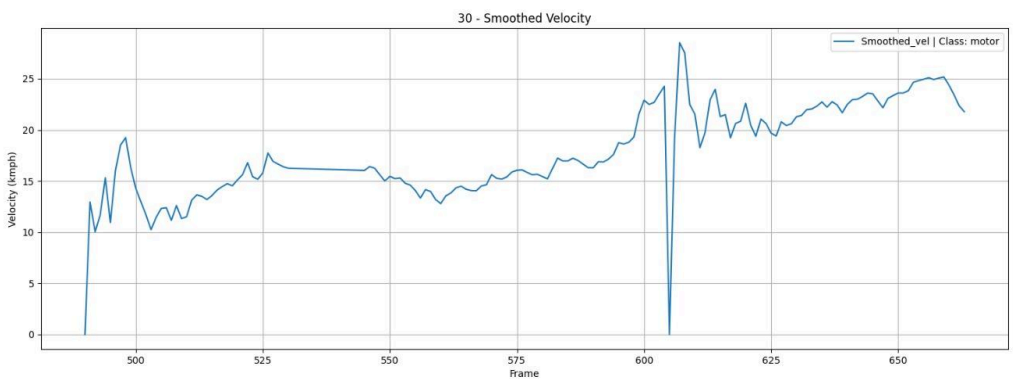

Fig. 2: Example of a hard stop at frame 610

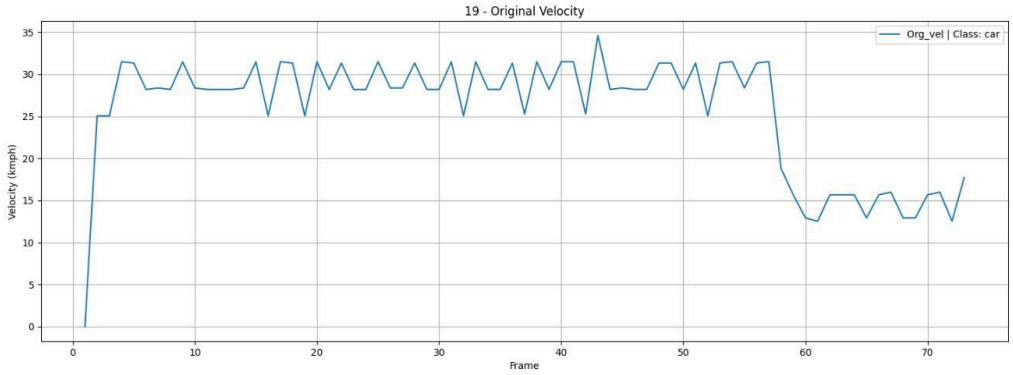

Fig. 3a

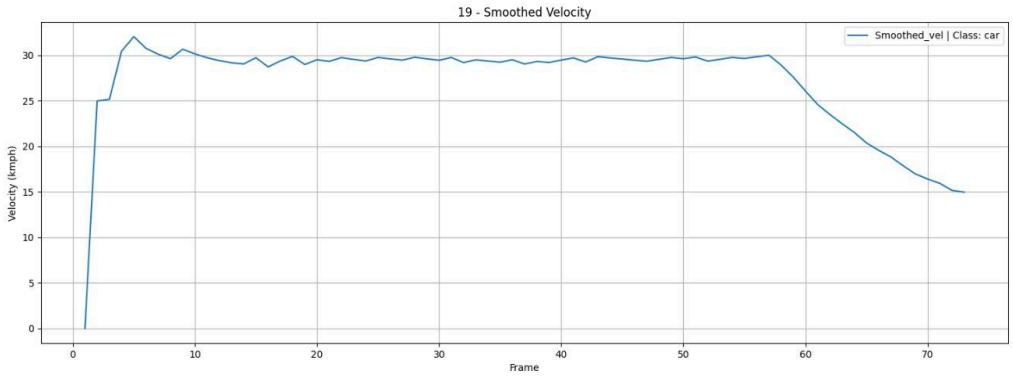

Fig. 3b

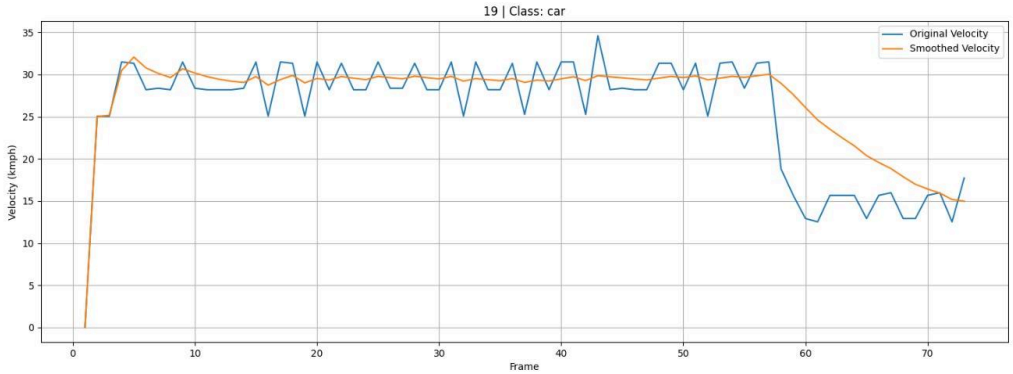

Fig. 3c

Fig. 3: Difference between original velocity vs smoothed velocity

3. Problems Faced

During this week, there were several challenges encountered while correcting the Oriented Bounding Box (OBB) annotations.

One major issue was the lack of clear understanding of the equation of angle or a proper reference in order to calculate the bounding boxes orientation angle of individual objects (vehicles) from the value in the dataset. Even though the dataset provided the parameters for the vehicle orientation, the method needed in order to construct the OBB was not clearly understood. Due to this confusion, forming OBB gave incorrect results such as a rotated bounding box which clearly did not align with the vehicle in the image.

Another issue was the size of the image we were provided with. The original image provided had a large resolution. The working image was reduced to approximately 1900 pixels. This made the bounding boxes to be placed at random places on the image.

To fix this problem the team had to find another way to do Object Bounding Box annotations and therefore worked with Axis-Aligned Bounding Box (AABB) for the time being. AABB is much easier because it does not require any rotation/angle calculations unlike OBB and thus allows us to correctly show the positions of the vehicles in the entire frame.

4. Agenda for Next Week

The tasks planned for next week will be dedicated to the further analysis and enhancement of the dataset that will enable better motion estimation and analysis.

The initial task will be the further data visualization regarding the specifics of various classes of vehicles available in the dataset. This can involve comparison of velocity pattern of the different classes of vehicles and studying of whether the motion of different types of vehicles has different behavior. With the help of such visualizations, patterns and trends will be identified that might prove to be beneficial in subsequent phases of the project.

Data cleaning and dataset refinement will also be an important task. In the course of the first analysis, the team noticed that not all labels of vehicle classes seem to be correct or fit well

together. Thus, the dataset will be checked and cleaned to rectify the misclassified data and eliminate any error or inconsistencies involved in data.

Finally, the team aims to start researching and choosing a suitable machine learning method to solve the stop detection problem. Using the knowledge acquired during the visualizations of the velocity and the cleanup dataset, various potential methodologies of categorization will be considered. It will aim at establishing an appropriate approach that will enable a proper differentiation between the various stopping behaviors, including hard stops and momentary stops, based on the available trajectory and velocity characteristics in the dataset.